

DATA ANALYSIS PROJECT REPORT

Temporal Patterns in Urban Crime

Project Title	Temporal Patterns in Urban Crime: A Time-and-Location Analysis of Los Angeles Crime Reports
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Course / Section	Programming in Python [Section D]
Instructor	Md. Tanzeem Rahat
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1. Executive Summary

This project looks at 134,198 crime incidents reported across the City of Los Angeles to see whether when and where a crime happens follows any kind of pattern. Using Pandas for cleaning and grouping and NumPy for the statistical side, the dataset was broken down by hour, day of the week, patrol area, and premises type. The results show that crime is not spread evenly through the day: the Afternoon window (12:00-16:59) accounts for the largest single share of incidents at about 27%, narrowly ahead of Night. A Z-score analysis of the 21 patrol areas identified Central, Southwest, and Pacific as statistically dense crime zones, with Central alone sitting nearly three standard deviations above the citywide average. The type of crime committed also shifts with the time of day vehicle theft is concentrated overnight, while shoplifting and petty theft cluster in the afternoon and about 30% of all incidents happen on a public street. A closer look at the data also surfaced a data-quality issue worth flagging: over 95% of all records are dated to 2024 alone, and report dates cluster far more tightly than occurrence dates, which likely inflates the number of statistical 'outlier' days.

2. Introduction and Project Objectives

Knowing when and where crime tends to happen is one of the more practical questions a city's police department has to deal with it feeds directly into how patrol shifts are staffed and which neighborhoods get more attention at which hours. This project uses a public LA crime dataset to look for exactly that kind of pattern, focusing on time of day, day of the week, and geographic area rather than trying to explain why any individual crime happened.

Item	Your Content
Dataset name	City of Los Angeles Crime Data (2020-2024 subset)
Dataset source	LA Open Data Portal Crime Data from 2020 to Present, data.lacity.org (dataset ID 2nrs-mtv8)
Main project goal	Determine whether the time a crime occurs, and where it occurs, follow a predictable pattern that could help prioritize patrol coverage.
Research Question 1	How does the frequency of reported crime vary across the time of day and the day of the week?
Research Question 2	Which patrol areas show a statistically significant concentration of crime relative to the citywide average (Z-score analysis)?

Research Question 3 (optional)	Does the mix of crime types committed change depending on the time of day?
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3. Dataset Description

Each row in the dataset represents a single reported crime incident in Los Angeles. The columns cover when it was reported and when it actually occurred, the patrol area and reporting district, the type of crime, the victim's sex (where recorded), the premises type, the weapon involved (where applicable), and a rough street level location. The data covers occurrences from January 2020 through December 2024.

3.1 Basic Dataset Information

Property	Value
Number of rows	134,198 incident records
Number of columns	11 original columns 17 after feature engineering
Data types present	Categorical (crime type, area, weapon, premises), date time (report/occurrence dates), and numeric (time, area code, district number)
Target or key variable(s)	CRIME, AREA NAME, TIME OF DAY (no single prediction target this is a descriptive/exploratory analysis)
Potential issues observed at first glance	Heavy missingness in WEAPON (81%) inconsistent/missing VICT SEX codes DATE RPTD clustered in a narrow window unlike DATE OCC

3.2 Initial Observations

The first thing that stood out was how uneven the missing data is across columns: WEAPON is blank for about 81% of records, which makes sense once you notice most of the top crime types (vehicle theft, burglary, shoplifting) don't typically involve a weapon in the first place. VICT SEX also had a mix of blanks and an 'X' placeholder that needed to be collapsed into a single missing-value category before it could be used for grouping. There's also a 131 category CRIME column and a 279 category PREMISES column, both heavily skewed toward a handful of common categories, so most of the analysis focuses on the top 5 of each rather than all of them.

4. Data Cleaning and Preparation

Four cleaning and preparation steps were applied before any analysis, aimed at making the categorical columns consistent and the date/time columns usable for grouping.

Issue Found	Action Taken	Reason / Justification	Code Reference
Possible missing DATE OCC values	Dropped rows where DATE OCC was NaN using <code>df[~df["DATE OCC"].isna()]</code>	The whole analysis is built around time, so a row with no occurrence date can't be used. In this particular file the check removed 0 rows, but it was kept in as a safeguard in case the CSV changes.	Cell 2

Inconsistent victim sex labels (mixed case, 'X' and blank entries)	Standardized casing with <code>.str.upper().str.strip()</code> , then mapped 'X' and 'UNKNOWN' to NaN	Keeps VICT SEX to a small, comparable set of categories (F / M / H) instead of a mix of placeholders that would otherwise get treated as separate groups.	Cell 2
Missing WEAPON values in about 81% of rows	Filled with the label "WEAPON UNKNOWN" instead of dropping those rows	Dropping four out of every five rows would have wiped out most of the dataset. Most of these are property crimes (theft, burglary) where no weapon is ever logged, so a blank isn't really 'missing data' in the usual sense - labelling it keeps the row for every other analysis.	Cell 2
Date and time columns stored as text / packed integers	Converted DATE RPTD and DATE OCC with <code>pd.to_datetime()</code> ; converted TIME OCC to numeric and derived HOUR OCC	The raw columns can't be grouped by hour, weekday, month, etc. until they are proper datetime/numeric types.	Cell 2

5. Feature Engineering

The raw dataset only gives a packed HHMM time value and a date string, so four new columns were derived to make time-based grouping possible.

New Feature	How It Was Created	Why It Is Useful
HOUR OCC	TIME OCC is stored as an HHMM integer (e.g. 1506); dividing by 100 and flooring it gives the hour	Lets crimes be grouped by hour instead of a four digit time code
TIME OF DAY	A small function buckets HOUR OCC into Morning (5-11), Afternoon (12-16), Evening (17-20) or Night (21-4)	Turns 24 individual hours into four shifts that are easier to compare and match how patrol shifts are usually described
DAY OF WEEK	Extracted from DATE OCC with <code>.dt.day_name()</code>	Needed to check whether crime is spread evenly across the week
DAY TYPE	<code>np.where()</code> flags Saturday/Sunday as 'Weekend' and everything else as 'Weekday'	Weekday vs weekend is a cleaner comparison than seven separate days, and matters for staffing decisions

6. Analysis Methodology

The project stays within Pandas, NumPy, and Matplotlib, as required:

- Pandas handled loading the CSV, cleaning, type conversion, and just about all the grouping/pivoting - by time of day, by area, by day of week, and combinations of these used throughout the analysis.

- NumPy was used for the actual statistics: mean and standard deviation for the area Z-score calculation in Section 7.2, and again for the daily-count outlier detection in Section 8.
- Matplotlib produced every chart in this report bar charts for the categorical comparisons, a stacked bar chart for the time vs crime type relationship, and a line chart with threshold markers for the outlier detection.

7. Analysis and Visualizations

7.1 Research Question 1: Temporal Distribution

How does the frequency of reported crime vary across the time of day and the day of the week?

Incidents were grouped into the four TIME OF DAY buckets, and separately by DAY OF WEEK and the derived DAY TYPE (weekday vs weekend) to see whether either dimension showed a clear pattern.

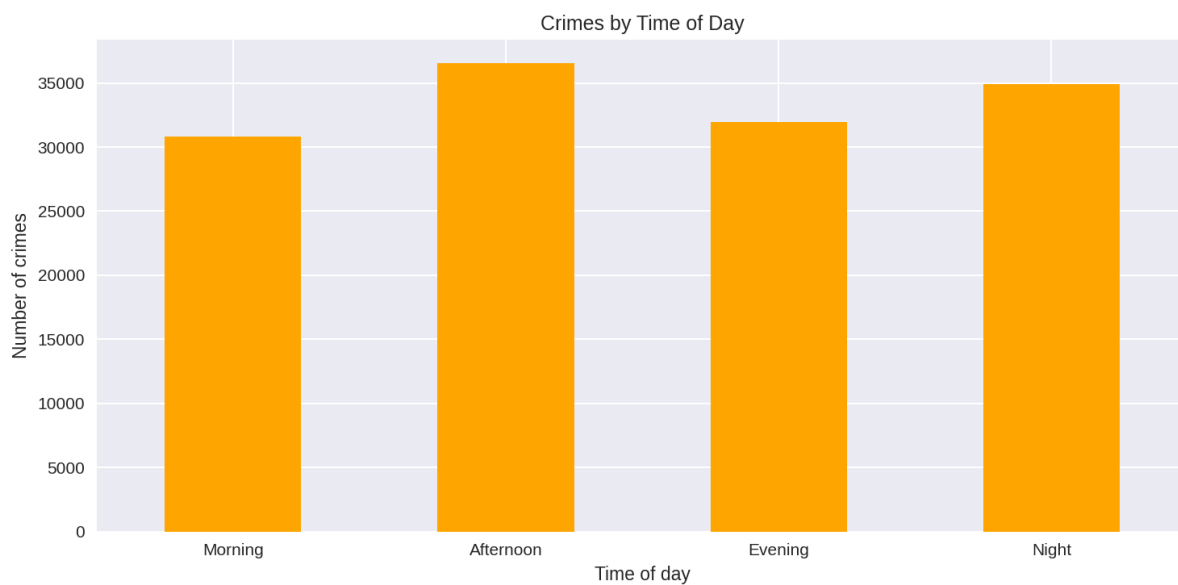


Figure 1. Total number of reported crimes by time of day.

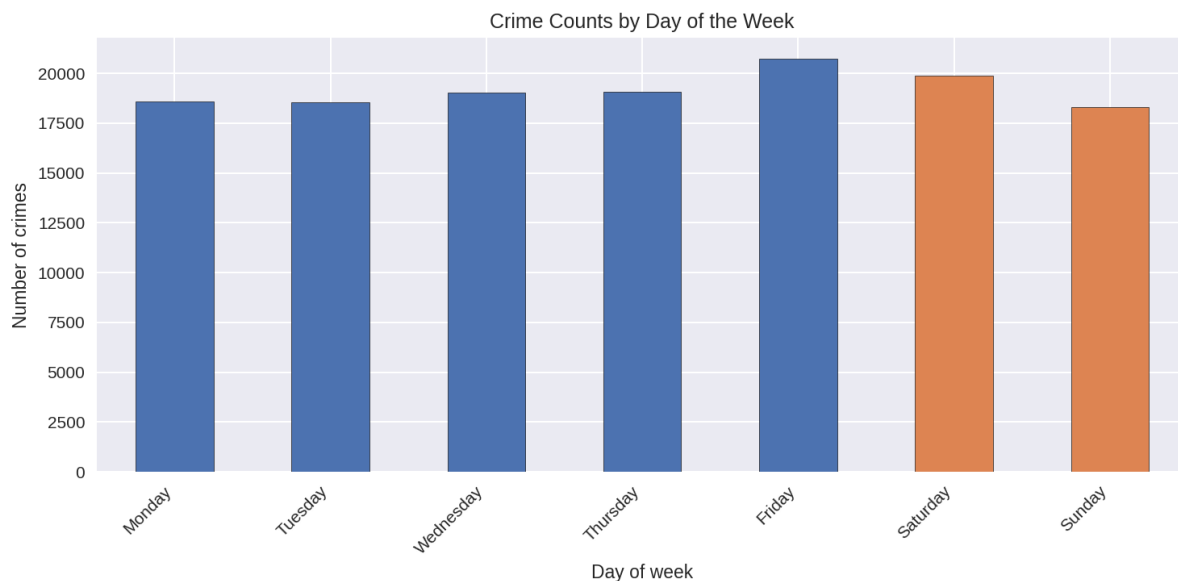


Figure 2. Total number of reported crimes by day of the week (Saturday and Sunday highlighted).

Afternoon incidents lead with 36,545 (about 27.2% of the total), just ahead of Night at 34,915 (26.0%); Evening and Morning trail at roughly 23.8% and 23.0%. The gap between the busiest and quietest time-of-day bucket is a real but fairly modest 5,700 incidents, so time of day matters, but it doesn't dominate the picture. Day of week tells a more interesting story once it's normalized: the dataset contains far more weekdays than weekend days, so raw totals make weekdays look busier, but on a per-day basis the average weekend day (about 102 incidents) actually edges out the average weekday (about 96). Friday is the single busiest day of the week overall.

7.2 Research Question 2: Geographic Hotspots

Which patrol areas show a statistically significant concentration of crime relative to the citywide average?

A Z-score was calculated for each of the 21 patrol areas using the citywide mean and standard deviation of incident counts per area, then filtered for areas more than 1.5 standard deviations above average.

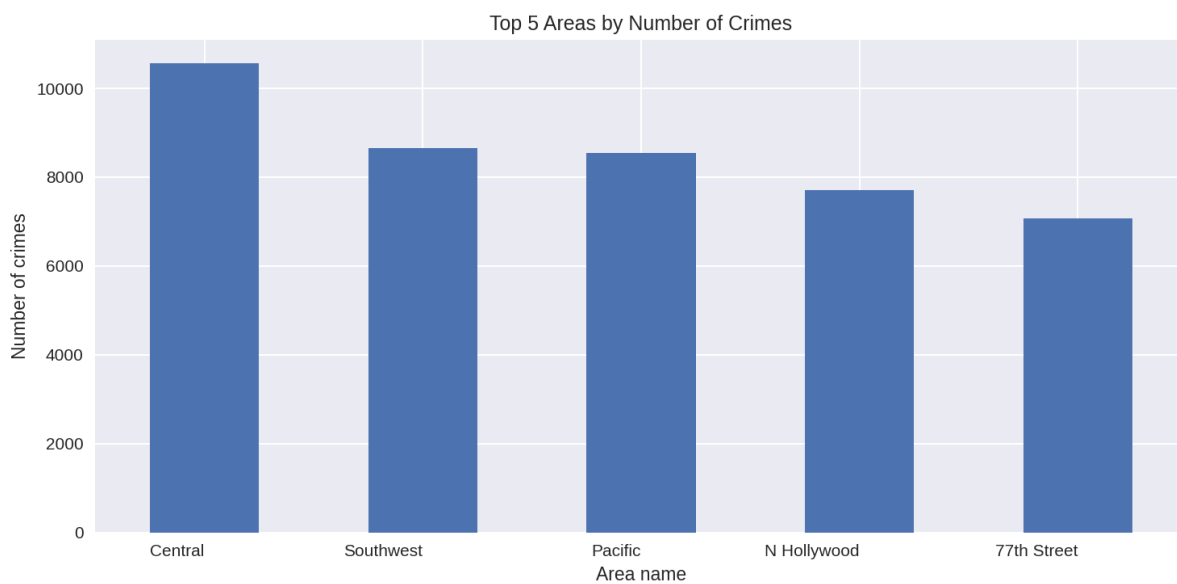


Figure 3. The five patrol areas with the highest number of reported crimes.

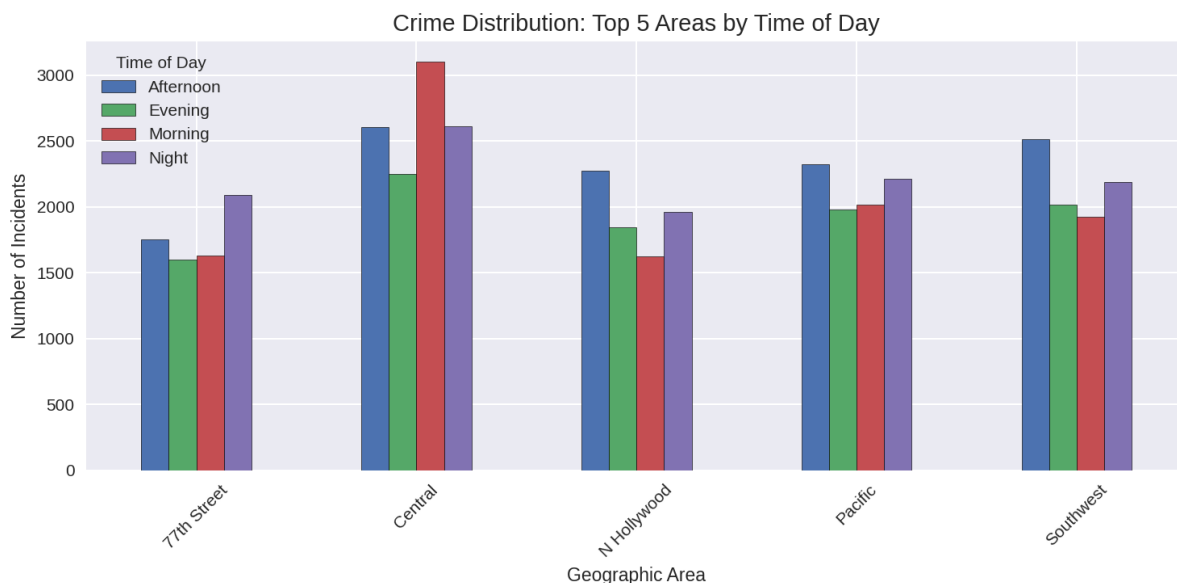


Figure 4. How those top five areas break down by time of day.

Only three areas actually clear the $Z > 1.5$ bar despite five showing up in the raw top-5 count: Central ($Z \approx 2.94$, more than 10,500 incidents against a citywide average of about 6,390), Southwest ($Z \approx 1.59$), and Pacific ($Z \approx 1.51$). N Hollywood and 77th Street round out the raw top five by volume but don't clear the statistical threshold a reminder that 'top 5 by count' and 'statistically unusual' aren't quite the same question. Central in particular stands out across every time of day bucket in Figure 4, not just at a specific hour.

7.3 Research Question 3: Relationship Analysis

Does the mix of crime types committed change depending on the time of day?

The five most common crime types were pivoted against TIME OF DAY to see whether particular offenses cluster at particular hours rather than being spread evenly.

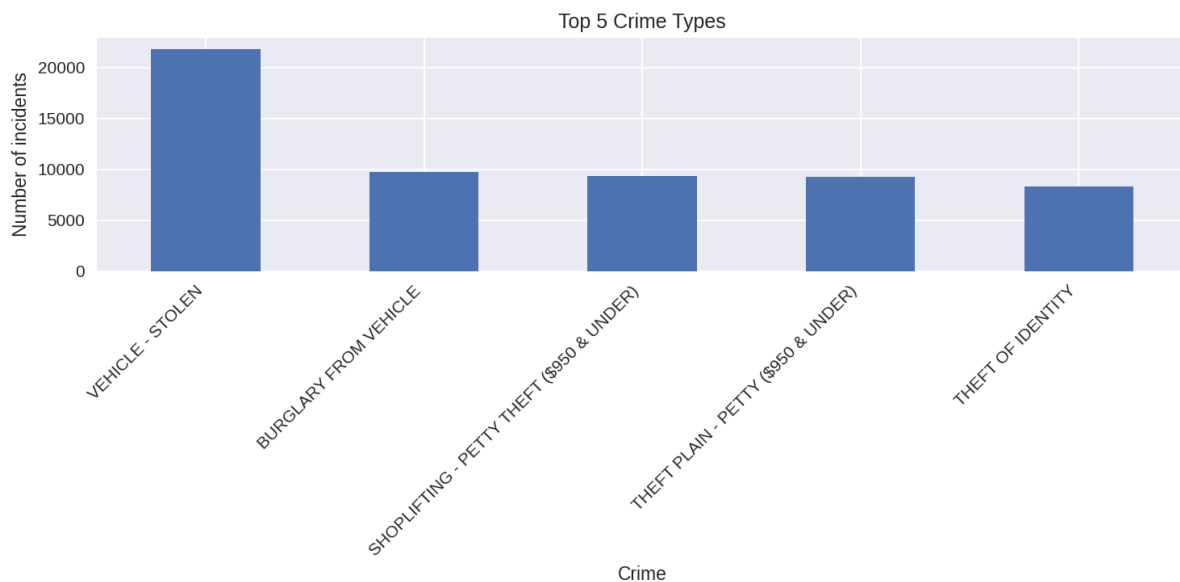


Figure 5. The five most frequently reported crime types overall.

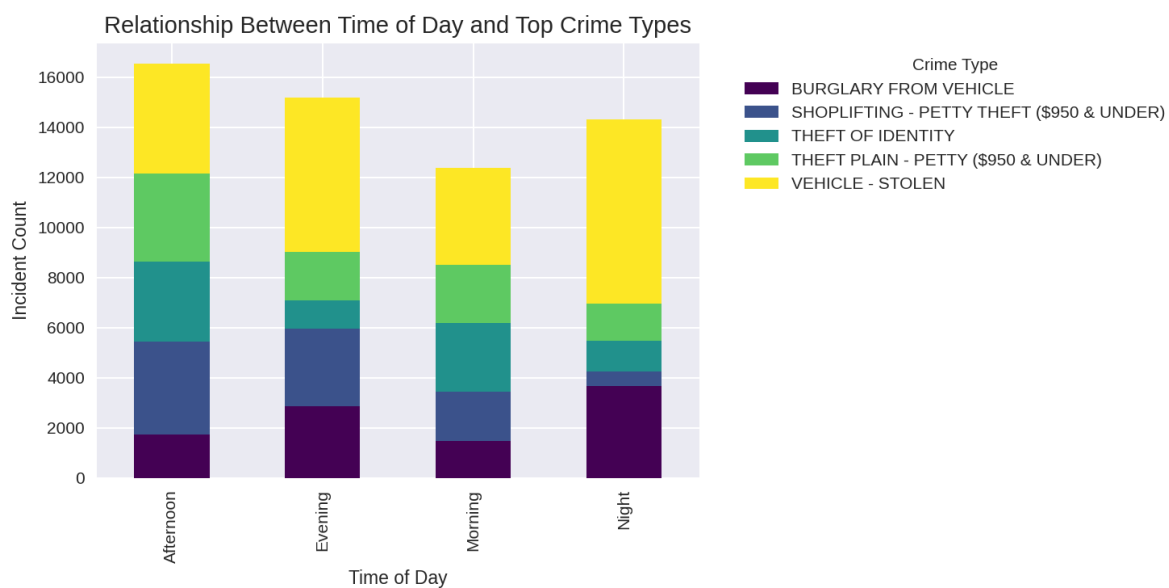


Figure 6. How the top five crime types are distributed across time of day.

Vehicle theft is the single most common crime type overall (21,823 incidents) and is heavily skewed toward Night (7,363, about a third of all vehicle theft cases) consistent with cars being stolen while parked and unattended overnight. Shoplifting shows almost the opposite pattern, concentrated in the Afternoon (3,688) and dropping sharply at Night (596), which lines up with store operating hours. Burglary from vehicle is fairly evenly spread but still leans toward Night and Evening. This confirms that time-of-day patterns aren't just a citywide average they look quite different depending on which crime type is being tracked.

7.4 Additional Analysis: Where Crimes Occur

PREMISES was one of the original columns but wasn't used anywhere in the first pass of this analysis, so it's worth a quick look on its own.

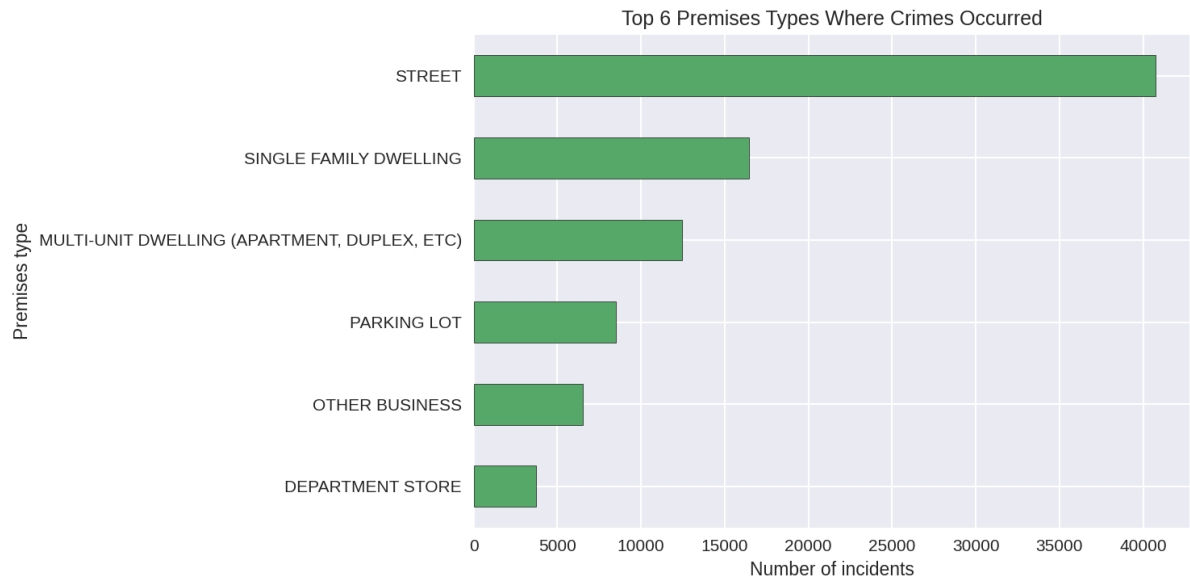


Figure 7. The six most common premises types where crimes were reported.

Public streets account for 40,744 incidents about 30% of everything in the dataset, more than double the next category (single-family dwellings at 16,446). Combined with parking lots (8,509), a clear majority of recorded crime happens in outdoor or semi public spaces rather than inside private homes or businesses, which lines up with vehicle-related crime (theft, burglary from vehicle) being the two most common offense types overall in Section 7.3.

8. NumPy-Based Custom Computation

Computation used	Daily crime-count outlier detection using a mean + 2*standard-deviation threshold (built on top of the same Z-score logic used for the area analysis in Section 7.2)
Purpose	To flag specific dates where the number of reported incidents was unusually high compared to a typical day, rather than relying on a visual guess from the chart
Formula or logic	<code>daily_counts = df.groupby("DATE OCC").size();</code> a day is an outlier if <code>daily_counts > daily_counts.mean() + 2 * daily_counts.std()</code>
Result / takeaway	The average day had about 97 reported incidents (std ~171), and 103 days crossed the ~439-incident threshold. Almost all of those 103 days fall in a tight window in early to mid 2024 rather than being spread across the full

2020-2024 range (see Figure 8) - which turned out to say more about when incidents were entered into the system than about an actual citywide crime spike, later confirmed by the year-level breakdown in Section 10 (Figure 9).

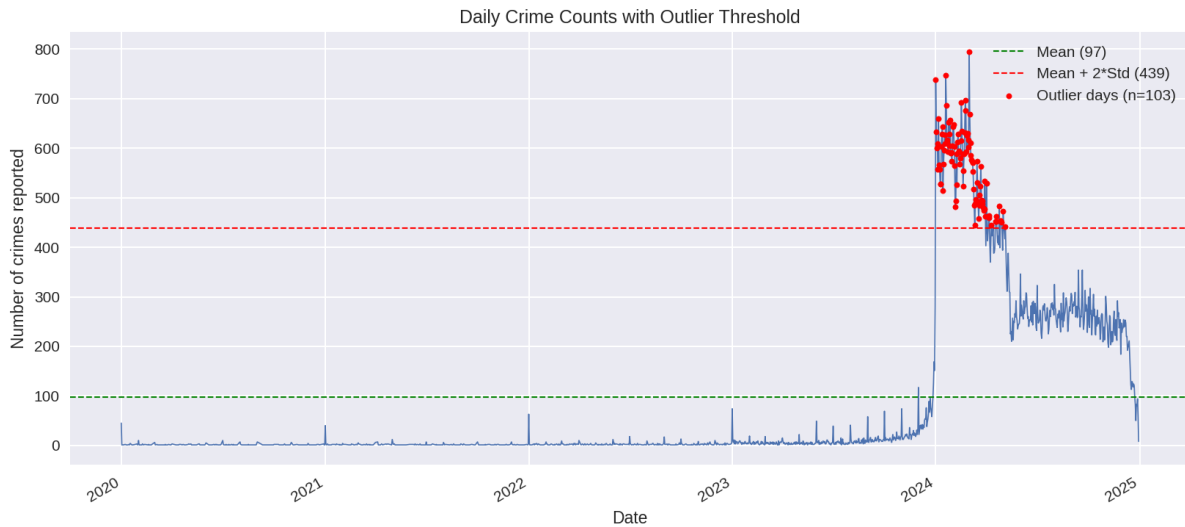


Figure 8. Daily reported crime counts, with the mean and the 2-standard-deviation outlier threshold marked.

9. Key Findings

1. Afternoon is the peak window for reported crime (36,545 incidents, ~27.2% of the total), but the spread across all four time-of-day buckets is fairly narrow time of day shapes the picture more than it dominates it.
2. Weekend days see a slightly higher average number of incidents per day (~102) than weekday days (~96), even though the dataset contains far more weekdays overall a pattern that only shows up once the totals are normalized by the number of days in each group.
3. Crime is not evenly spread across patrol areas: Central's Z-score of 2.94 puts it well above every other area, at roughly 65% more incidents than the citywide area average, while only two other areas (Southwest, Pacific) clear the 1.5 statistical threshold despite five areas appearing in the raw top-5 list.
4. Vehicle theft and shoplifting sit at opposite ends of the clock - vehicle theft peaks at Night (a third of all cases) while shoplifting peaks in the Afternoon and nearly disappears at Night, suggesting the two crime types call for different patrol timing.
5. The daily-outlier check flagged 103 unusually high-count days, but almost all of them fall in a narrow early-2024 window rather than being spread across the full 2020-2024 range - which lines up with LAPD's March 2024 switch to a new records system (DATE RPTD is bunched into 2024-2025) rather than a genuine citywide crime spike.
6. That reporting-artifact suspicion holds up at the year level too: over 95% of all 134,198 records are dated to 2024 alone, with 2020-2023 combined making up less than 5% - so this dataset is best treated as a detailed 2024 snapshot rather than an even five-year sample.
7. Crime location is heavily concentrated in public space: about 30% of all incidents (40,744) happened on a street, more than double the next largest category (single family dwellings), which tracks with vehicle-related offenses being the most common crime types in the dataset.

10. Limitations

A few things limit how far these results should be trusted. The biggest one is a year-level imbalance that the daily outlier check in Section 8 only hinted at: when incidents are counted by occurrence year, over 95% of every record in this dataset is dated to 2024, with 2020-2023 combined making up less than 5% (Figure 10). That is consistent with LAPD's March 2024 transition to a new records-management system - it looks like most pre-2024 cases simply weren't carried over. In practice this means the dataset is closer to a detailed snapshot of 2024 than a genuine five-year trend, and both the outlier day detection in Section 8 and any year over year comparison should be read with that in mind rather than treated as evidence of a real citywide crime spike.

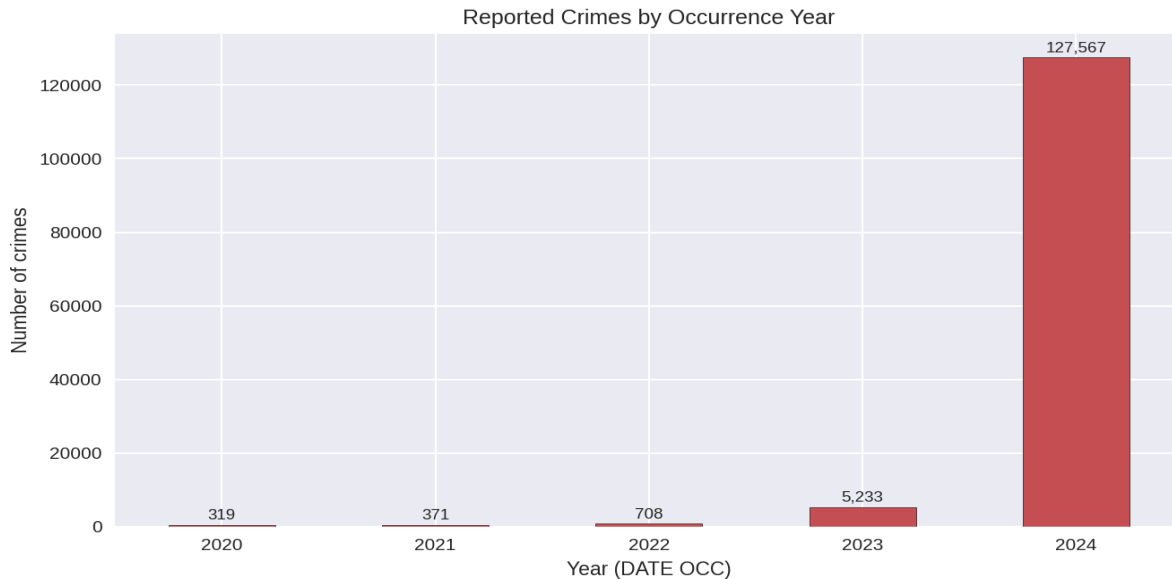


Figure 9. Reported crimes by occurrence year - the overwhelming majority are dated to 2024.

Second, WEAPON is missing for about 81% of rows and VICT SEX for roughly 37% after cleaning, and there's no way to tell from the data alone whether those are truly 'no weapon involved' cases or simply unrecorded - the WEAPON UNKNOWN label used here treats both the same way. Third, every pattern reported here is a correlation, not a cause: knowing that vehicle theft peaks at night says nothing about why it does, and area-level differences in reported crime may reflect differences in patrolling or reporting behavior as much as differences in actual crime. Finally, this is a single city's data covering one dominant year, so none of these patterns should be assumed to generalize elsewhere or over time.

11. Conclusion

Working through this dataset shows that crime in Los Angeles is not randomly distributed across time or space - certain hours, certain days, certain areas, certain premises types, and certain crime types are all statistically over-represented in fairly specific ways. Afternoon and Night both stand out as higher volume windows, Central division stands out geographically by a wide margin, most incidents happen in outdoor or semi public spaces rather than inside homes, and the type of crime being committed clearly interacts with when it happens. At the same time, digging into the outlier detection - and then following up with a year level breakdown surfaced a data-quality issue that a purely surface-level read of the chart would have missed: this dataset is really a detailed picture of 2024 rather than an even five-year sample. That was a useful reminder to check the assumptions behind a statistic rather than just report the number. With more time, this analysis could be extended by joining in additional context, such as population or patrol staffing per area, to see whether the geographic hotspots identified in Section 7.2 hold up once they're normalized by something other than raw incident count, or by revisiting the year over year comparison once more consistently reported years of data become available.

12. References / Dataset Source

City of Los Angeles Crime Data from 2020 to Present. LA Open Data Portal, [data.lacity.org](https://data.lacity.org/Public-Safety/Crime-Data-from-2020-to-Present/2nrs-mtv8). Dataset ID 2nrs-mtv8. <https://data.lacity.org/Public-Safety/Crime-Data-from-2020-to-Present/2nrs-mtv8>

13. Appendix

One additional pattern came up during cleaning that didn't fit neatly under any of the three research questions above, but is worth including since it supports Key Finding relating to weapon involvement.

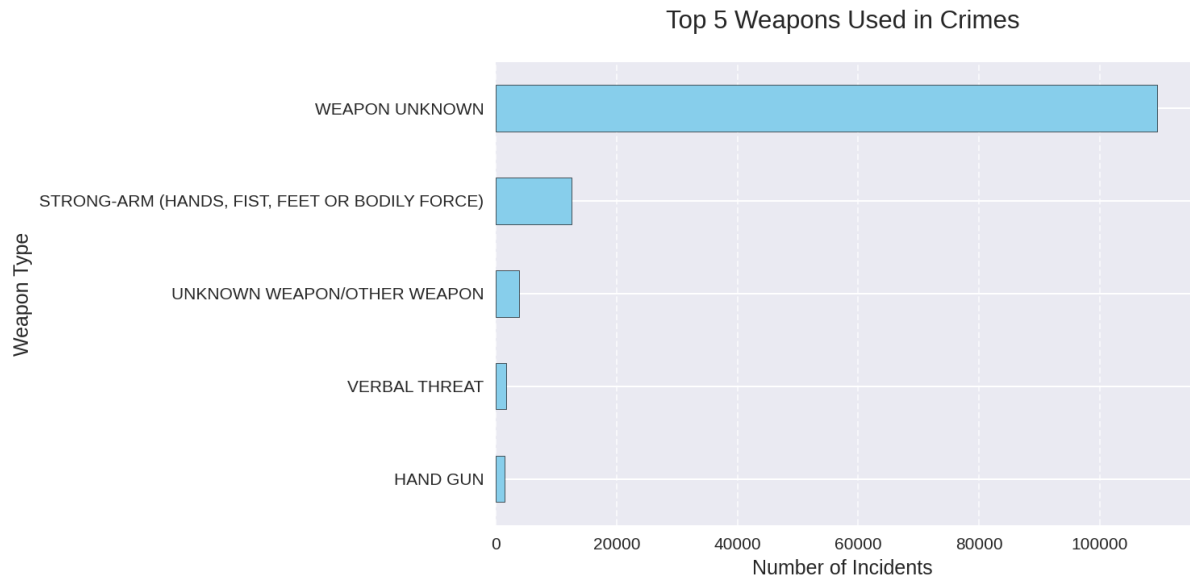


Figure 10. The five most common weapon categories recorded (WEAPON UNKNOWN dominates due to the missingness discussed in Section 3.2 and Section 10).

Excluding the WEAPON UNKNOWN placeholder, 'strong-arm' (hands, fists, feet, or bodily force) is by far the most common weapon category logged at 12,600 incidents well ahead of firearms, which appear in only about 1.1% of all records. This fits with the crime mix identified in Section 7.3: property crimes like vehicle theft and shoplifting make up most of the dataset, and those rarely involve a weapon at all.